

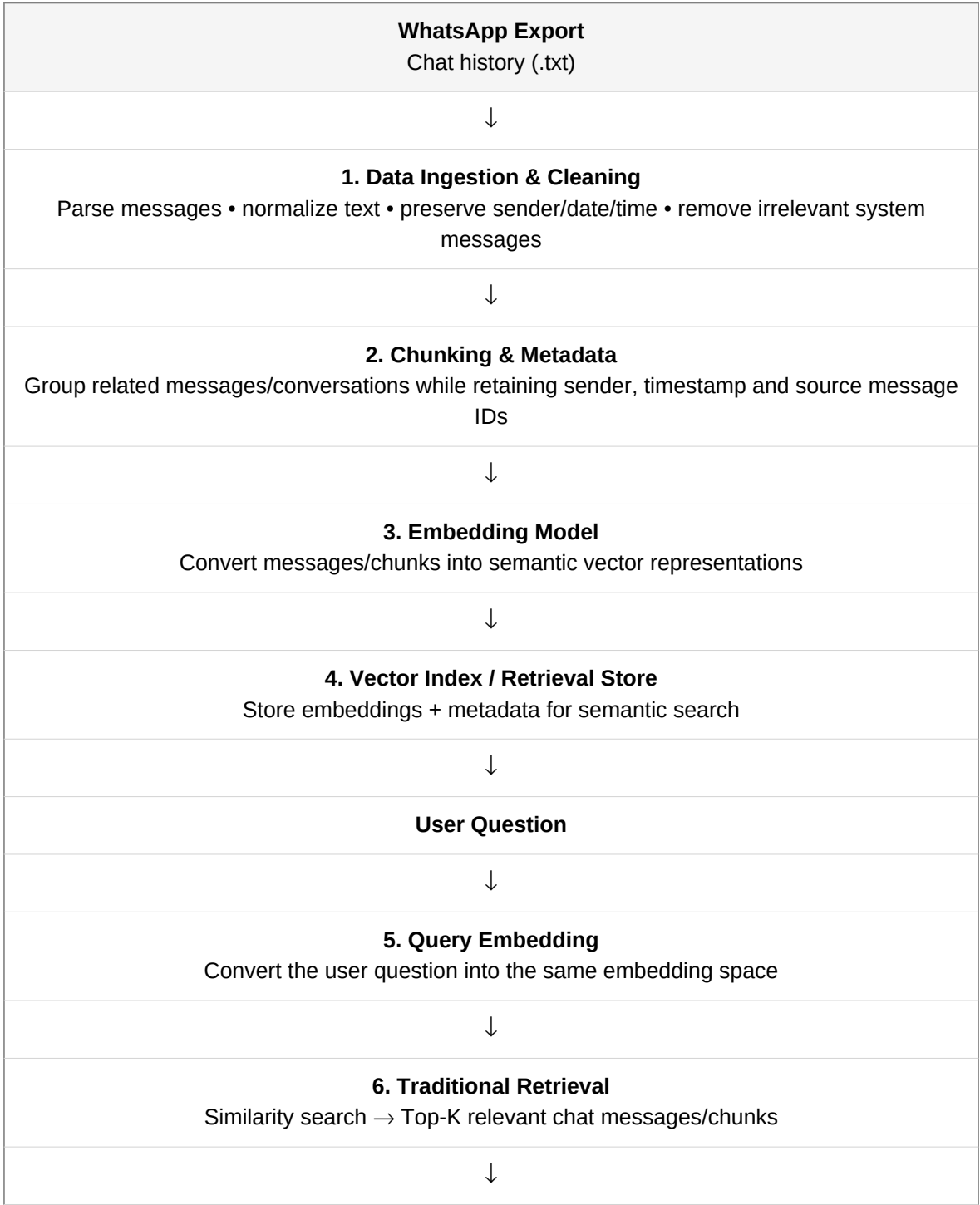
WhatsApp Group Chat Q&A; Chatbot

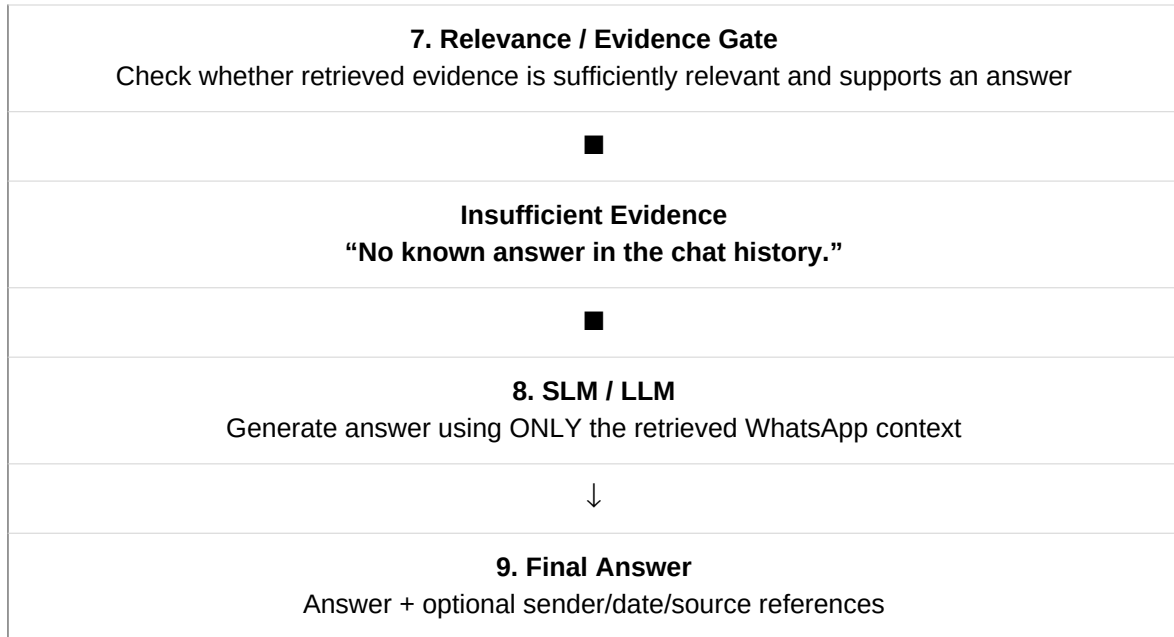
Final Architecture — Embedding + Traditional Retrieval + SLM/LLM

1. Objective

Build a question-answering chatbot whose knowledge is strictly limited to a WhatsApp group chat history. The chatbot should answer only when relevant evidence exists in the chat history; otherwise it must return “No known answer in the chat history.”

2. Final Architecture





3. Component Responsibilities

Component	Purpose
WhatsApp History	The single source of truth / knowledge base.
Data Processing	Clean and structure raw WhatsApp export without changing factual content.
Embedding Model	Enable semantic matching between user questions and historical messages.
Vector Index	Efficiently retrieve the most semantically relevant historical content.
Relevance Gate	Prevent unsupported questions from reaching the generation stage.
SLM / LLM	Understand retrieved context and formulate a natural-language answer.
Answer Layer	Return the answer and, where useful, supporting message metadata.

4. Strict Knowledge Boundary

The SLM/LLM is **not** the knowledge source. It is only the language-generation layer. The system should provide the model with retrieved WhatsApp evidence and explicitly constrain it to that evidence.

Decision rule: If the retrieval/evidence layer cannot establish that the answer is present in the chat history, do not generate an answer from general model knowledge. Return the fixed fallback response.

5. Example Flow

User Question	When is the deployment scheduled?
Retrieved History	Rahul: The deployment is scheduled for Friday at 6 PM.
Evidence Gate	Sufficient evidence found
SLM/LLM	Generates a concise answer from the retrieved message

Final Answer	The deployment is scheduled for Friday at 6 PM.
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Unknown question example: If the user asks something that cannot be supported by the retrieved chat history, the system returns: **“No known answer in the chat history.”**

6. Recommended Design Principle

Retrieve first → validate evidence → generate second. This gives substantially better control than relying on an LLM prompt alone to prevent outside knowledge.

The architecture is model-agnostic: the embedding model, vector store, and SLM/LLM can be replaced independently without changing the overall design.